

Project Walkthrough — Nuclear Plant Leak Detection

This walkthrough summarizes the mining tasks performed, the models used, the 80/20 train-test split, and the results with their associated visualizations.

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Data & Preprocessing

- Dataset: 14 days at 1-minute frequency (20,160 rows).
- Preprocessing includes event-aware outlier handling, gap treatment (≤ 5 min ffill/bfill, ≤ 30 min interpolation, > 30 min flagged), denoising (rolling median + Savitzky-Golay), and PCA (95% variance, 14 components).
- Visuals: EDA (data\mining_results\eda_comprehensive.png), Processing Overview (data\processed\processing_overview.png), PCA panels (data\mining_results\dimensionality_reduction.png).

Mining Task 1 — Multi-class Classification

Task: Multi-class risk classification (Normal, Elevated, Medium Risk, High Risk).

Split: 80/20 train-test with stratification. CV: 5-fold on training set.

- Decision Tree: Accuracy=1.000, CV_Mean=1.000, CV_Std=0.000
- Random Forest: Accuracy=1.000, CV_Mean=1.000, CV_Std=0.000
- Logistic Regression: Accuracy=1.000, CV_Mean=1.000, CV_Std=0.000
- SVM: Accuracy=1.000, CV_Mean=1.000, CV_Std=0.000
- KNN: Accuracy=1.000, CV_Mean=1.000, CV_Std=0.000

Visuals: Confusion Matrix (data\mining_results\confusion_matrix_random_forest.png), Feature Importance (data\mining_results\feature_importance.png), Summary (data\mining_results\model_comparison_summary.png).

Deduction: Perfect separation on synthetic data; radiation sensors and sump flow dominate importance.

Mining Task 2 — Binary Classification

Task: Binary leak detection (Leak vs No-Leak).

Split: 80/20 train-test with stratification.

- Decision Tree: Accuracy=1.000, AUC=1.000
- Random Forest: Accuracy=1.000, AUC=1.000
- Logistic Regression: Accuracy=1.000, AUC=1.000
- SVM: Accuracy=1.000, AUC=1.000
- KNN: Accuracy=1.000, AUC=1.000

Visuals: ROC (Random Forest) (data\mining_results\roc_visualization_random_forest.png), Summary (data\mining_results\model_comparison_summary.png).

Deduction: $AUC \approx 1.00$ across models; features provide perfect separability in this synthetic setup.

Mining Task 3 — Regression (Leak Rate)

Task: Regression of leak rate (LPM) on leak-only subset.

Split: 80/20 train-test.

- Linear Regression: $R^2=0.980$, RMSE=2.891
- Ridge Regression: $R^2=0.979$, RMSE=2.959
- Random Forest: $R^2=0.997$, RMSE=1.062
- SVR: $R^2=0.921$, RMSE=5.705
- KNN: $R^2=0.985$, RMSE=2.461

Visuals: Regression Scatter (data\mining_results\regression_scatter.png), Summary (data\mining_results\model_comparison_summary.png).

Deduction: Random Forest achieves $R^2 \approx 0.997$; slight spread only at highest leak rates.

Mining Task 4 — Association Rules

Task: Mine co-occurrence patterns using Apriori on binned numeric features.

Result: No interesting rules found at $\text{support} \geq 0.01$ and $\text{lift} \geq 1.1$.

Deduction: Leak events are sparse with diverse states; lowering thresholds or focusing on precursor subsets could surface rules.